

Mathematics and Statistic

Assessment Schedule – 2023

Mathematics and Statistics 91261: Apply algebraic methods in solving problems

Evidence Statement

NOTE: Timing and extent of rounding/truncating may lead to different values/conclusions. Apply consistency.

Question	Evidence	Achievement (u) Apply algebraic methods in solving problems.	Merit (r) Apply algebraic methods, using relational thinking, in solving problems.	Excellence (t) Apply algebraic methods, using extended abstract thinking, in solving problems.
(a)	$(3x - 2)(5x + 3) = 0$ $15x^2 - x - 6 = 0$	Correct equation stated Accept non-integer co-efficients N1 – if not an equation ie not set =0		
(b)	$15ap + 6aq - 5bp - 2bq$ $= 3a(5p + 2q) - b(5p + 2q)$ $= (3a - b)(5p + 2q)$	Correct factorisation.		
(c)	$\frac{(3x + 2)(4x - 3)}{(3x + 2)(x - 6)} = -2$ $\frac{4x - 3}{x - 6} = -2$ $4x - 3 = -2x + 12$ $6x = 15$ $x = \frac{5}{2}$ or 2.5 Must reject solution $x = -\frac{2}{3}$ OR $\frac{12x^2 - x - 6}{3x^2 - 16x - 12} = -2$ $12x^2 - x - 6 = -2(3x^2 - 16x - 12)$ $12x^2 - x - 6 = -6x^2 + 32x + 24$ $18x^2 - 33x - 30 = 0$ $3(6x^2 - 11x - 10) = 0$ $3(2x - 5)(3x + 2) = 0$ **line $x = \frac{5}{2}$ or $-\frac{2}{3}$ $x = -\frac{2}{3}$ makes the denominator undefined so reject. Only answer is $x = \frac{5}{2}$	Correct factorisation top and bottom. OR second method Correct Factorisation **line Accept if 3 not factored out as solutions can still be found. N1 – correct factorisation of either numerator or denominator N1 – correct expansion and set =0 ie $18x^2 - 33x - 30 = 0$	Correct solution. Only one, students MUST reject $x = -\frac{2}{3}$	
(d) (i)	$P(t) = 1740000 \times 1.0112^t$ In 2023, $t = 13$ $P(13) = 1740000 \times 1.0112^{13}$ $P(13) = 2011087.842$ Population in 2023 will be approximately 2012000, or just over 2 million.	Formed a successful model for $P(t)$. Accept in millions 3 and 1.74	Correct approximate value found. Accept 2.012 million	
(d) (ii)	$P(t) = 3000000, \quad t = 78$ $3000000 = 1740000 \times r^{78}$ $\left(\frac{3000000}{1740000}\right) = r^{78}$	Set up correct equation to solve. Accept in millions 3 and 1.74	Has made some progress towards answer, has found an equation for either $\log r$ or r .	Correct value of r found. $r = 1.007$ Alternative answer for r without using logs is

	$78 \log r = \log \left(\frac{3000000}{1740000} \right)$ $\log r = \frac{\log \left(\frac{3000000}{1740000} \right)}{78}$ $r = 10^{\frac{\log \left(\frac{3000000}{1740000} \right)}{78}}$ $r = 1.007008$ This is a growth rate of approximately 0.7%. Alternate working $3000000 = 1740000 \times r^{78}$ $\left(\frac{3000000}{1740000} \right) = r^{78}$ *** $r = \left(\frac{3000000}{1740000} \right)^{\frac{1}{78}}$ $r = 1.007008$		Alternate method: given line *** Or equivalent	$r = \sqrt[78]{\frac{50}{29}}$ Students must interpret r (see next page)
(e)	$\log x = 2 \log (kx)$ $\log x = \log(kx)^2$ $x = k^2 x^2$ $k^2 x^2 - x = 0$ $x(k^2 x - 1) = 0$ $x = 0 \quad \text{or}$ $x = \frac{1}{k^2} \quad k \neq 0$	Simplifies expression = one correct log rule and one correct algebra step OR 2 correct log rules N1: line 1	Simplifies and gets $k^2 x^2$ removes log function and attempts solution	Correctly states valid answer. Working shown Restrictions on k not needed BUT discuss with students!

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	1 of N1	1 of u	2 of u	3 of u	1 of r	2 of r	1 of t	2 of t

Question	Evidence	Achievement (u) Apply algebraic methods in solving problems.	Merit (r) Apply algebraic methods, using relational thinking, in solving problems.	Excellence (t) Apply algebraic methods, using extended abstract thinking, in solving problems.
TWO (a)	$\frac{4a^{\frac{5}{4}}}{\sqrt[3]{64a^{12}}}$ $= \frac{4a^{\frac{5}{4}}}{4a^4}$ $= a^{-\frac{11}{4}}$ $= \frac{1}{a^{\frac{11}{4}}} \quad \text{or} \quad \frac{1}{\sqrt[4]{a^{11}}}$	Correct response. N1: removes the 4 from both numerator and denominator		
(b)	$9x^2 + 25y^2 = 225$ $3x + 5y = k$ $x = \frac{k-5y}{3}$ $(k-5y)^2 + 25y^2 = 225$ $50y^2 - 10ky + k^2 - 225 = 0$ $a = 50 \quad b = -10k \quad c = k^2 - 225$ $b^2 - 4ac = 0$ $-100k^2 = -45000$ $k^2 = 450$ $k = \pm 21.21$ so $k > 21.21$ or $k < -21.21$ to prevent the drain crossing the garden.	Rearrange elliptical equation and Make x the subject of the linear equation Show some progress to solving simultaneous equations	Substitute for x into elliptical equation and simplify.	Find discriminant Solve for k. And Interpret answer in context
(c)	$\frac{12}{(x+4)(x-4)} + \frac{3}{2(x+4)}$ $= \frac{24 + 3(x-4)}{2(x+4)(x-4)}$ $= \frac{24 + 3x - 12}{2(x+4)(x-4)}$ $= \frac{3x + 12}{2(x+4)(x-4)}$ $= \frac{3(x+4)}{2(x+4)(x-4)}$ $= \frac{3}{2(x-4)}$ $x \neq 4$ or -4 Alternate method: $= \frac{24(x+4) + 3(x-4)(x+4)}{2(x+4)(x-4)(x+4)}$	Lowest common denominator found. OR Common denominator found -cubic see alternate method N1: does not simplify numerator	Correct answer is found. Restrictions on values of x not required.	
(d)	$(2^4)^m = 2^n$ $2^{4m} = 2^n$ $n = 4m$	Correct simplified answer is found.		
(e)	Volume of cuboid	Sets up 2 volumes	Factorised and	Found and

	$V = x(x + 5)(x + 6)$ $= x(x^2 + 11x + 30)$ $V = x^3 + 11x^2 + 30x$ Volume of cube = $(2x)^3 = 8x^3$ Volume cube = volume of cuboid $8x^3 = x^3 + 11x^2 + 30x$ $7x^3 - 11x^2 - 30x = 0$ $x(7x^2 - 11x - 30) = 0$ $x(7x + 10)(x - 3) = 0$ $x = 0, 3, \text{ or } -\frac{10}{7}$ $x = 3$ due to contextual constraints Volume = $8x^3 = 8 \times 3^3$ $= 216 \text{ cm}^3$	as equal. Simplified cubic equation found. <i>N1: Gives equation of cube $(2x)^3 = 8x^3$</i> <i>N1: correct expansion of cuboid $= x^3 + 11x^2 + 30x$</i>	solved, giving 3 solutions for x. OR Only 2 solutions given and not flagged lost 0	communicated volume, eliminating incorrect values of x. Units required
--	---	---	--	---

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	1 of N1	1 of u	2 of u	3 of u	1 of r	2 of r	1 of t	2 of t

Question	Evidence	Achievement (u) Apply algebraic methods in solving problems.	Merit (r) Apply algebraic methods, using relational thinking, in solving problems.	Excellence (t) Apply algebraic methods, using extended abstract thinking, in solving problems.
THREE (a) (i)	$x^{-2} = \frac{1}{16}$ $x^2 = 16$ $x = \pm 4$ $x = 4 \text{ is the only valid solution}$	Gives $x = \pm 4$ <i>N1: finds</i> $x^2 = 16$ <i>N1: finds $x = 4$ only but no indication of -4 also being a solution</i>	Rejects $x = -4$ As cannot have negative base Answer $x = 4$	
(a) (ii)	$= \log(x^2 y^2) + \log(6) - \log(3y)$ $= \log\left(\frac{6x^2 y^2}{3y}\right)$ $= \log(2x^2 y)$	Correct solution. <i>N1: gets to line 2</i>		
(b)	Let $\log_5 x = y$ $6y^2 + 9y - 6 = 0$ $2y^2 + 3y - 2 = 0$ $(2y - 1)(y + 2) = 0$ $y = \frac{1}{2} \text{ or } y = -2$ So, $\log_5 x = \frac{1}{2}$ $x = 5^{\frac{1}{2}} = \sqrt{5} = 2.236$ NB only accept $+\sqrt{5}$ or $\log_5 x = -2$ $x = 5^{-2} = \frac{1}{25} = 0.04$	Forms factorised equation. <i>N1: subs in variable</i>	Uses algebraic methods to get one correct solution.	Uses algebraic methods to get both correct solutions.
(c)	Let the roots be n and $7n$. Equation is $(x - n)(x - 7n)$ $7n^2 = k$ and $-8n = 16$ $n = -2$, $7n^2 = 28$, $k = 28$	Factored form of roots or similar, algebraically shown. OR CAO (no working)	Value of k is found.	
(d) (i)	If $a^3 + b^3 = 27x^3 + 8$ Then $a = 3x$ and $b = 2$ $27x^3 + 8 = (3x + 2)(9x^2 - 6x + 4)$	One bracket correct. <i>N1: misidentifying cube roots but set up consistently</i>	Correct factored form.	
(d) (ii)	$\frac{x^3 + 8}{x^2 - 4} = k$. Rewriting gives $\frac{(x+2)(x^2 - 2x + 4)}{(x+2)(x-2)} = k, \quad x \neq 2 \text{ Line 1}$ $\frac{x^2 - 2x + 4}{x - 2} = k \quad \text{Line 2}$ $x^2 - 2x + 4 = kx - 2k$ $x^2 - 2x - kx + 4 + 2k = 0$ $x^2 - (2 + k)x + (4 + 2k) = 0$ For two solutions, $b^2 - 4ac > 0$ $(-(2 + k))^2 - 4(4 + 2k) > 0$ $k^2 - 4k - 12 > 0$	Simplified rational expression is found - Line 2. <i>N1: sets up line 1</i>	Correct use of discriminant and quadratic for k is found.	Correct ranges for k are provided.

	$k < -2$ and $k > 6$			
--	----------------------	--	--	--

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	1 of N1	1 of u	2 of u	3 of u	1 of r	2 of r	1 of t	2 of t

Cut Score:

The minimum cut scores that were required at each grade in the 2022 paper were:

	Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
Minimum Score	0 - 6	7 - 12	13 - 18	19 - 24